

Uncertainty-Conditioned Reinforcement Learning with Evidential Policy Networks for Autonomous Systems

Antonio Galdes
University of Malta
Faculty of ICT
antonio.galdes.22@um.edu.mt

ABSTRACT

Autonomous parking systems trained with reinforcement learning typically assume reliable localisation, yet sensor quality fluctuates unpredictably in practice. This report presents an uncertainty-conditioned reinforcement learning architecture that addresses this limitation through two complementary mechanisms. The first mechanism embeds the full posterior covariance of an Extended Kalman Filter (EKF) localisation pipeline, comprising nine elements including cross-covariance terms, directly into the policy observation vector. The second mechanism replaces the standard Gaussian actor head with a Normal-Inverse-Gamma evidential output layer, yielding per-action epistemic and aleatoric uncertainty estimates in a single forward pass. A Proximal Policy Optimisation agent is trained within the CARLA simulator under dynamically varying RTK-GNSS fix-state conditions modelled by a discrete-time Markov chain. A 2×2 factorial ablation study is proposed to isolate the contribution of each uncertainty layer to collision avoidance under degraded localisation. The observation space was designed to exclude simulator-specific information, such that all dimensions correspond to quantities reproducible outside the simulation environment.

Categories and Subject Descriptors

I.2.6 [Artificial Intelligence]: Learning—*connectionism and neural nets*; I.2.9 [Artificial Intelligence]: Robotics—*autonomous vehicles*

General Terms

Algorithms, Design, Reliability

Keywords

Reinforcement learning, evidential deep learning, uncertainty quantification, autonomous parking, localisation uncertainty, Extended Kalman Filter

1. INTRODUCTION

Reinforcement learning (RL) applied to vehicle control typically assumes that the localisation system provides reliable

pose estimates throughout operation [1]. In practice, this assumption does not hold. RTK-GNSS accuracy degrades from centimetre-level to metre-level depending on satellite visibility, multipath reflections and atmospheric conditions [2], and a parking system that collides with obstacles when localisation quality deteriorates cannot be deployed under automotive functional safety standards such as ISO 26262 [3]. Policies trained without awareness of localisation quality are therefore inherently brittle.

Two distinct types of uncertainty contribute to this brittleness. Epistemic uncertainty reflects the model’s ignorance of unfamiliar states and can be reduced with additional training data; aleatoric uncertainty captures irreducible noise in the environment and persists regardless of data quantity [4, 5].

Existing RL approaches address these two concerns in isolation, and no prior work has attempted to combine them within a single policy. Evidential deep learning (EDL) methods have been applied to critic networks [6], to perception modules feeding model predictive control [7] and to value-based discrete-action agents [8]. Separately, policies have been conditioned on localisation uncertainty without evidential outputs [9], and uncertainty has been employed for active exploration rather than adaptive control [10]. However, no evidential output head has been placed on the actor of a continuous-action policy-gradient agent, nor have raw EKF covariance elements been fed as RL observation dimensions for vehicle control. The question is whether combining both mechanisms within a single architecture produces measurably safer behaviour under degraded conditions. Both gaps remain open.

An architecture that addresses these gaps is presented in this report. The proposed system is trained in the CARLA driving simulator [11] on an outdoor parking task under dynamically varying GNSS quality, implemented as a three-container Docker stack. The aims and objectives are stated in Section 2; the relevant literature is reviewed in Section 3; the proposed solution is described in Section 4; and the evaluation plan is outlined in Section 5.

2. AIMS AND OBJECTIVES

The aim of this research is to develop and evaluate an uncertainty-conditioned RL architecture that enables an autonomous parking agent to degrade gracefully under deteriorating localisation quality. The following objectives have been defined:

1. Implement an EKF-based localisation pipeline within CARLA that fuses RTK-GNSS and IMU data via the

`robot_localization` ROS 2 package [12] and produces a posterior covariance matrix reflecting sensor degradation.

2. Develop an evidential Proximal Policy Optimisation (PPO) algorithm [13] in which the actor network outputs Normal-Inverse-Gamma (NIG) distributions over continuous actions [14], enabling single-pass quantification of epistemic and aleatoric uncertainty.
3. Conduct a 2×2 ablation study comparing the full method against baselines lacking one or both uncertainty layers, under both nominal and degraded GNSS conditions.

3. BACKGROUND AND LITERATURE REVIEW

3.1 Reinforcement Learning for Vehicle Control

Sequential decision-making is formulated by reinforcement learning as a Markov Decision Process (MDP) defined by a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, P the transition function, R the reward function and $\gamma \in [0, 1]$ the discount factor [15, 2]. A policy $\pi_\theta(a|s)$ maps states to action distributions, parameterised by neural network weights θ , and the objective is to determine θ such that the expected discounted return $\mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r_t]$ is maximised.

On-policy learning is stabilised by Proximal Policy Optimisation (PPO) [13] through a clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right] \quad (1)$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ denotes the probability ratio and $\hat{A}_t$ is the advantage estimate. Advantages are computed using Generalised Advantage Estimation (GAE), which interpolates between Monte Carlo and one-step temporal-difference estimates via a trace-decay parameter λ [13]. The clipping parameter ϵ constrains the magnitude of policy updates, thereby preventing the large destabilising steps that characterise unconstrained methods such as Trust Region Policy Optimisation (TRPO) [16]. PPO is trained by optimising a combined loss comprising the clipped policy loss, a value function loss and optionally an entropy bonus. The value function loss regresses the critic's state-value estimate $V_\theta(s_t)$ toward the empirical return $\hat{R}_t$:

$$L^{\text{VF}}(\theta) = \left(V_\theta(s_t) - \hat{R}_t \right)^2. \quad (2)$$

An approximate Kullback-Leibler (KL) divergence between the updated and previous policies may additionally be monitored; if this divergence exceeds a configurable threshold, the optimisation epoch is terminated early to prevent excessive policy change [13]. Observation normalisation via running statistics is employed as standard practice to stabilise training when observation dimensions span different scales [17]. The interaction between these components provides the foundation upon which the evidential extensions described in Section 4 are built.

3.2 Evidential Deep Learning

Evidential deep learning places a prior over the likelihood parameters rather than learning point estimates. For regression, a Normal-Inverse-Gamma (NIG) prior is imposed over the mean μ and variance σ^2 of a Gaussian likelihood [14], and the network outputs four parameters $(\gamma, \nu, \alpha, \beta)$ per output dimension, from which two uncertainty types are derived:

$$\sigma_{\text{epi}}^2 = \frac{\beta}{\alpha - 1}, \quad \sigma_{\text{alea}}^2 = \frac{\beta}{\nu(\alpha - 1)}. \quad (3)$$

Sensoy et al. [18] introduced EDL for classification using Dirichlet priors, and Amini et al. [14] subsequently extended it to continuous regression via the NIG formulation. A survey of EDL methods is provided by Gao et al. [19]. Oh and Shin [20] investigated the gradient shrinkage problem inherent in the NIG loss and proposed multi-task learning strategies to improve prediction accuracy without sacrificing calibration quality.

In contrast to these single-pass evidential approaches, ensemble-based alternatives [21, 22] require multiple forward passes to obtain uncertainty estimates. Monte Carlo Dropout [23] provides an alternative single-model approximation but still necessitates multiple stochastic passes at inference time. A single NIG forward pass produces both uncertainty types simultaneously, rendering it particularly suitable for control policies operating at high frequencies where inference latency is constrained. This computational property motivates the adoption of the NIG formulation for the actor head described in Section 4.

3.3 Uncertainty Estimation in Reinforcement Learning

Several recent works have integrated uncertainty estimation into RL agents, and these can be grouped by where the uncertainty mechanism is placed within the agent architecture. On the critic side, Akgül et al. [6] place a NIG evidential prior on the value function $V(s)$ to address non-stationarity, whilst the actor remains a standard Gaussian policy. Stutts et al. [8] similarly apply evidential learning to Q-value estimation in a discrete-action DQN framework.

Theoretical and risk-oriented approaches have also been explored. Charpentier et al. [5] provide a foundation for disentangling epistemic and aleatoric uncertainty in the RL setting, and Clements et al. [24] estimate risk in model-free agents through distributional value functions.

Ensemble-based methods represent an alternative family of approaches. Kim et al. [25] condition the policy on ensemble disagreement for robustness in unfamiliar MuJoCo scenarios, Zhang et al. [22] employ ensemble-based epistemic uncertainty for safe RL in autonomous driving, and Hoel et al. [26] apply ensemble uncertainty to tactical decision-making. On the perception side, Ham and Ahn [7] employ NIG-parameterised EDL within distributionally robust MPC constraints in CARLA, coupling evidential uncertainty with model predictive control rather than with a learned policy.

Puthumanaim et al. [9] present the most conceptually similar input-uncertainty mechanism to the one proposed here: task-specific uncertainty maps are constructed from localisation quality and fed to a Soft Actor-Critic (SAC) agent [27]. However, processed uncertainty abstractions are employed rather than raw covariance elements, and no evidential method is used for output uncertainty. A gap is therefore identified: no evidential NIG head has been placed on the actor of a continuous-action policy-gradient agent,

and the full EKF posterior covariance, including off-diagonal cross-covariance terms, has not been exposed as explicit observation dimensions for ego-vehicle control.

3.4 Extended Kalman Filter Localisation

The Extended Kalman Filter (EKF) is a recursive Bayesian estimator that maintains a Gaussian belief over the vehicle state $\mathbf{x}_t = [x, y, \psi]^\top$ [2]. The prediction step inflates the covariance according to process noise $\mathbf{Q}$:

$$\mathbf{P}_{t|t-1} = \mathbf{F}_t \mathbf{P}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}, \quad (4)$$

where $\mathbf{F}_t$ is the state transition Jacobian, $\mathbf{P}_{t-1|t-1}$ is the posterior covariance from the previous step and $\mathbf{Q}$ is the process noise covariance. The update step then incorporates a measurement with noise covariance $\mathbf{R}$:

$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t) \mathbf{P}_{t|t-1}, \quad (5)$$

where $\mathbf{K}_t$ is the Kalman gain, $\mathbf{H}_t$ is the measurement Jacobian and $\mathbf{I}$ is the identity matrix. Between GNSS fixes, the covariance $\mathbf{P}_{t|t}$ grows; when a fix arrives, it collapses, producing a characteristic sawtooth profile. The `robot_localization` ROS 2 package [12] provides the EKF node used in this work. Kovacs et al. [28] demonstrated that process noise configuration significantly affects both EKF convergence rate and localisation accuracy, a finding that informed the IMU noise parameterisation adopted in the proposed system.

3.5 Autonomous Parking with Reinforcement Learning

RL-based approaches to autonomous parking have been explored due to the difficulty of hand-engineering controllers that generalise across diverse lot geometries. Yang et al. [29] proposed an offline RL framework using SAC-N with ensemble-based Q-value penalisation, representing the closest existing work to uncertainty-aware parking RL; however, ensemble disagreement on the critic is employed rather than evidential methods on the actor, and localisation uncertainty is not incorporated. Codevilla et al. [30] established offline evaluation methodologies for driving models in CARLA, demonstrating the importance of systematic evaluation under varied conditions. The evaluation plan described in Section 5 draws on this methodology.

4. PROPOSED SOLUTION

4.1 System Architecture and Containerisation

The system is deployed as three Docker containers orchestrated via Docker Compose, illustrated in Fig. 1. The `carla-server` container runs CARLA 0.9.16 [11] headlessly with NVIDIA GPU passthrough, providing the physics engine and sensor models. The `ros2-bridge` container runs ROS 2 Jazzy and hosts both the CARLA-ROS bridge and the `robot_localization` EKF node [12] that fuses GNSS and IMU data. A custom `GnssNoiseRelayNode` intercepts GNSS messages and injects tier-dependent covariance before the EKF processes them. The `training` container is built on an NVIDIA NGC PyTorch base with Stable-Baselines3 [17] and subscribes to EKF output over the DDS middleware; a Gymnasium-compatible environment wraps the CARLA interface and constructs the observation vector from the fused sensor data.

This containerised separation ensures that EKF covariance arises naturally from sensor fusion rather than being artificially injected.

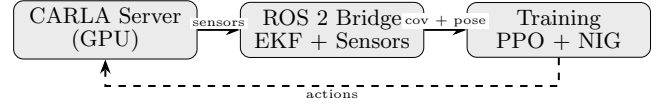


Figure 1: Three-container Docker architecture.

Each training episode loads one of several pre-computed parking lot floor plans from YAML layout files, and additional layouts can be generated for training or out-of-distribution (OOD) evaluation as needed. Bay occupancy and dynamic actor placement are randomised per episode. Periodic checkpoints are saved, and dedicated evaluation containers permit checkpoint playback through either a 2D bird’s-eye visualiser or windowed CARLA.

4.2 Layout Generation and Inspection

Parking lot geometry is defined by layout modules, each specifying a floor plan shape in a local coordinate frame. Three shapes have been implemented to date: a 60×45 m axis-aligned rectangle, a trapezoid (front width 48 m, rear width 30 m, depth 44 m) and a nine-sided irregular polygon. The first two are employed during training; the third is held out for OOD evaluation. Further layouts can be added by implementing additional modules following the same interface.

Each module defines bay positions (perpendicular, angled and parallel types following German EAR 05 dimensions), spawn transforms, patrol waypoints, pedestrian zone boundaries and perimeter cone positions interpolated along the polygon edges. A world-frame transform is then applied by an orchestrator script, which writes both YAML layout files consumed by the training environment and bird’s-eye PNG plots for visual verification (Fig. 2).

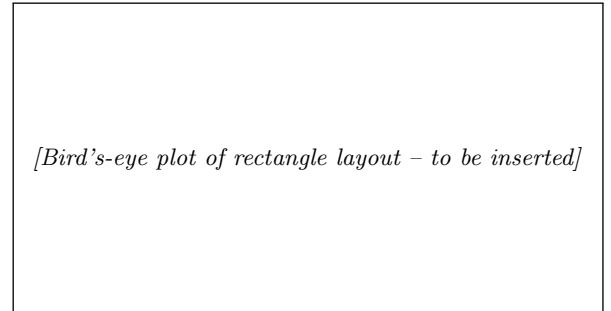


Figure 2: Generated bird’s-eye view of the rectangle floor plan showing bay positions (colour-coded by type), spawn points, patrol path and pedestrian zones.

A suite of CARLA inspection scripts permits visual verification of the generated layouts before training commences. A layout inspection mode spawns the full parking lot in windowed CARLA with debug overlays: bay outlines colour-coded by type, spawn points, pedestrian zone outlines and patrol waypoint paths. Additionally, a sensor inspection mode draws labelled dots at each sensor mount position and field-of-view arcs at the sensor range distances (Fig. 3),

whilst a live inspection mode spawns a 2D LiDAR and renders its scan output as debug dots in the CARLA world. All inspection modes run in dedicated Docker containers.

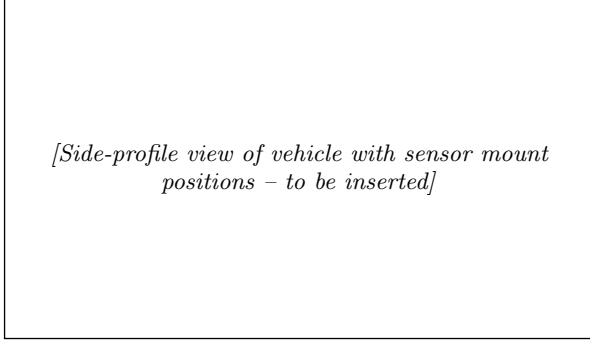


Figure 3: Sensor inspector side-profile view indicating IMU, GNSS antenna and 2D LiDAR mount positions on the vehicle body.

A live 2D bird’s-eye visualiser built with Pygame renders lot boundaries, bay outlines, NPC positions and the ego vehicle trajectory trail by tailing a per-step JSONL file written by the training environment (Fig. 4). The visualiser is fully detachable via a signal file.

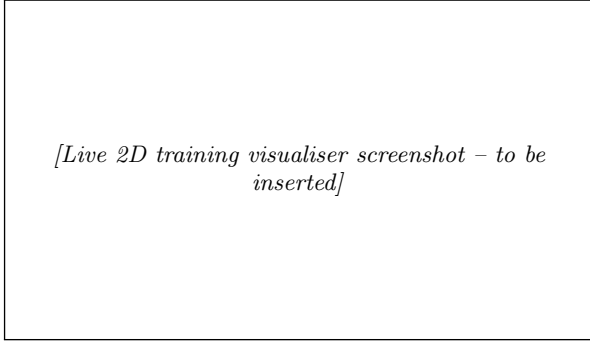


Figure 4: Live bird’s-eye training visualiser showing ego trajectory, bay outlines and NPC vehicles.

4.3 Observation Space

The observation vector comprises 20 dimensions divided into four semantic groups (Table 1). Absolute EKF position is excluded because it accumulates in the odometry frame across episodes; CARLA ground truth is employed only for reward computation.

Table 1: Observation space structure (20 dimensions).

Indices	Features	Source
0–2	$v_x, v_y, \dot{\psi}$	EKF velocity
3–11	$\sigma_x, \sigma_y, \sigma_\psi, P_{xx}, P_{yy}, P_{\psi\psi}, P_{xy}, P_{x\psi}, P_{y\psi}$	EKF covariance (log1p-transformed)
12–14	$dx, dy, d\psi$	Relative target
15–19	left/right/fwd clearance	2D LiDAR

The nine covariance features include both diagonal standard deviations and off-diagonal cross-covariance terms, and

a log1p transformation is applied to compress the heavy tails that arise under high-uncertainty conditions. Ablation flags control dimensionality: disabling covariance removes indices 3–11, yielding an 11-dimensional observation; disabling obstacle features removes indices 15–19.

4.4 Two-Layer Uncertainty Architecture

The proposed architecture combines two complementary uncertainty mechanisms, illustrated in Fig. 5. The input layer, consisting of EKF covariance elements in the observation vector, captures the confidence of the localisation system. The output layer, consisting of evidential NIG parameters on the actor, captures the confidence of the policy in its own action. Each layer responds to a distinct failure mode: degraded GNSS raises input uncertainty, whilst unfamiliar lot geometry raises output epistemic uncertainty. Their combination enables graduated responses ranging from cautious driving to full safety handoff.

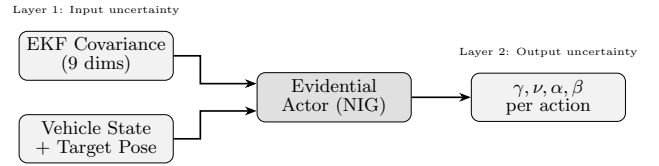


Figure 5: Two-layer uncertainty architecture.

4.5 Evidential Actor Head

The actor’s final layer produces NIG parameters $(\gamma, \nu, \alpha, \beta)$ per action dimension. Positivity is enforced through the softplus function, $\text{softplus}(x) = \log(1 + e^x)$, with offsets:

$$\begin{aligned}\nu &= \text{softplus}(z_\nu) + \epsilon \\ \alpha &= \text{softplus}(z_\alpha) + 1 \\ \beta &= \text{softplus}(z_\beta) + \epsilon\end{aligned}\quad (6)$$

where z_ν, z_α and z_β are the raw pre-activation outputs of the final network layer and $\epsilon = 10^{-6}$. The offset of 1 on α ensures finite variance in Eq. (3). All three parameters are clamped to a maximum of 100 to prevent divergence in the RL setting, where no ground-truth action target exists [20]. For action sampling, a Gaussian approximation is employed: $a \sim \mathcal{N}(\gamma, \sigma_{\text{alea}})$. The evidential regularisation loss takes the form of a prior-anchoring log-penalty:

$$\mathcal{L}_{\text{reg}} = \lambda \left[\overline{\log\left(\frac{\nu}{\nu_0} + 1\right)} + \overline{\log\left(\frac{\alpha}{\alpha_0} + 1\right)} \right] \quad (7)$$

where ν_0, α_0 are initialisation-time values, λ is linearly annealed from zero over a warmup period and $\overline{(\cdot)}$ denotes the mean over the action dimensions. This formulation differs from the supervised regression regulariser of Amini et al. [14] and is adapted for RL, where the residual $|a - \gamma|$ reflects policy stochasticity rather than prediction error. The total PPO loss is consequently $L = L^{\text{CLIP}} + c_v L^{\text{VF}} + \mathcal{L}_{\text{reg}}$, where L^{VF} is the value function loss from Eq. (2) and c_v is a weighting coefficient. All hyperparameter values (including λ, c_v , the learning rate and the PPO clipping range) are to be determined through Optuna-based hyperparameter optimisation using a Tree-structured Parzen Estimator (TPE) sampler prior to the ablation study.

Algorithm 1 presents the training loop. The **EvidentialPPO** subclass augments the standard PPO loss with the regularisation term from Eq. (7), and checkpoints are saved periodically, storing both the model weights and the observation normalisation statistics to ensure consistent scaling upon resumption or evaluation.

Algorithm 1 Evidential PPO training loop.

```

1: Load configuration from YAML
2: Initialise CARLA environment with EKF pipeline
3: Normalise observations via running statistics
4: Initialise EvidentialPPO with NIG actor head
5: for each rollout of  $n$  environment steps do
6:   Collect transitions  $(s_t, a_t, r_t, s_{t+1})$ 
7:   Cache NIG parameters  $(\gamma, \nu, \alpha, \beta)$ 
8:   Compute advantages via GAE
9:   for each optimisation epoch do
10:    Sample mini-batch
11:    Compute  $L^{\text{CLIP}}, L^{\text{VF}}$  and  $\mathcal{L}_{\text{reg}}$ 
12:    Update policy weights via combined loss
13:    if KL divergence exceeds threshold then
14:      Early-stop epoch loop
15:    end if
16:  end for
17:  Log uncertainty metrics to TensorBoard
18: end for
19: Save final model and normalisation statistics

```

A dual-encoder variant of the actor, termed the **UncertaintyConditionedActor**, processes vehicle state features and covariance features through separate encoder pathways before merging via concatenation into a shared trunk. This architectural separation permits the covariance pathway to develop specialised representations for uncertainty-dependent behaviour modulation.

4.6 Localisation Strategy

Scan-matching localisation against perimeter cones using Google Cartographer [31] was initially evaluated as the primary localisation source. This approach was abandoned because parked vehicles occlude the cones when bay occupancy is high, causing localisation to degrade in precisely the conditions where reliable operation is most needed. Wheel odometry was similarly rejected because the simulator provides zero-noise ground truth, which suppresses the covariance variation that the policy must learn to interpret. RTK-GNSS was therefore adopted, providing well-characterised degradation modes from centimetre-level accuracy (RTK fixed) through metre-level (standalone). This choice restricts the system to outdoor parking lots with sky visibility, which is acknowledged as a scope limitation in Section 5.

4.7 GNSS Noise Model

RTK-GNSS accuracy is modelled as four discrete tiers, and a discrete-time Markov chain at 20 Hz governs transitions between adjacent tiers:

$$\mathbf{T} = \begin{bmatrix} 0.995 & 0.005 & 0 & 0 \\ 0.003 & 0.992 & 0.005 & 0 \\ 0 & 0.004 & 0.993 & 0.003 \\ 0 & 0 & 0.005 & 0.995 \end{bmatrix} \quad (8)$$

where rows correspond to fixed (~ 2 cm), float (~ 30 cm), standalone (~ 2 m) and degraded (~ 5 m). No direct transitions

between non-adjacent tiers are permitted. The stationary distribution approximates $\pi \approx [0.39, 0.31, 0.22, 0.08]$, and mean dwell times of six to ten seconds produce five to twenty transitions per episode. Realistic IMU process noise derived from the VectorNav VN-100 datasheet [28] is configured in the EKF to ensure that covariance inflates appropriately between GNSS fixes.

4.8 Reward Function and Action Space

The initial reward function employs potential-based shaping [32] to guide the agent toward the target bay: $r_t = (d_{t-1} - d_t)/d_{\text{max}} - c_{\text{step}}$, with terminal rewards for success (+10), collision (−10) and out-of-bounds (−5). Success requires position error < 0.5 m, orientation error < 10° and speed < 0.1 m/s. The action space comprises steering $\in [-1, 1]$ and longitudinal control $\in [-1, 1]$. Both the reward structure and the action mapping are subject to revision based on initial training results.

4.9 Observation Design Rationale

All 20 observation dimensions were selected to avoid dependence on simulator-specific state. Table 2 summarises the sensor roles and data flow. Sensor mount positions and calibration parameters are externalised into YAML configuration files, permitting the same policy architecture to be evaluated under different configurations without modification.

Table 2: Sensor roles and data flow.

Sensor	Role	Feeds
RTK-GNSS	Position fixes	EKF update
IMU	Orientation	EKF prediction
2D LiDAR	Obstacle clearance	Obs. 15–19

5. EVALUATION PLAN

A 2×2 factorial ablation is planned to isolate the contribution of each uncertainty layer (Table 3). Each configuration is to be trained across three random seeds following hyperparameter optimisation. Evaluation episodes are to be conducted under several GNSS noise conditions, including individual fixed tiers and dynamic Markov switching; the specific conditions and episode counts may be adjusted based on preliminary results.

Table 3: Ablation study design.

	No EKF cov. (11d)	EKF cov. (20d)
Standard actor	<code>vanilla_ppo</code>	<code>input_unc</code>
Evidential actor	<code>output_unc</code>	<code>full_method</code>

The ablation cells correspond to the closest approaches in the literature. The `input_unc` configuration, which feeds localisation uncertainty to a standard actor, is analogous to the approach of Puthumanaim et al. [9]. The `output_unc` configuration, which places an evidential head on the actor without covariance input, mirrors the evidential-in-RL strategy of Akgül et al. [6], transposed from critic to actor. The `full_method` configuration combines both layers and has no direct counterpart in existing work.

The following metrics are currently planned:

1. **Success rate** under nominal and degraded GNSS, measuring whether the agent can park when localisation is reliable and how performance degrades as quality deteriorates.
2. **Collision rate under degraded GNSS**, the primary safety metric, aligned with offline evaluation practices for driving models [30].
3. **False attempt rate** under conditions too degraded for safe parking, quantifying how often each configuration attempts to park when it should not.
4. **Epistemic uncertainty on unseen layouts**, measuring OOD detection capability on lot geometries not encountered during training.
5. **Aleatoric uncertainty under RTK float**, measuring sensitivity to noisy outcomes and calibration of data uncertainty.
6. **Graceful degradation curve** of success rate against GNSS noise magnitude, showing the drop-off profile per ablation cell.

Each metric is to be averaged across three random seeds and reported with standard deviation. The specific statistical tests and presentation format will be determined once initial results are available.

The EKF pipeline is evaluated separately by comparing pose estimates against CARLA ground truth under each tier [28]. Root mean square error is computed for position and heading, and the EKF covariance is verified to vary appropriately with GNSS quality.

The central hypothesis is that neither uncertainty layer alone is sufficient for safe behaviour under degraded localisation; the full method should exhibit both the lowest collision rate and the most gradual degradation curve. The factorial structure is designed to reveal whether this holds through direct pairwise comparison.

5.1 Dynamic Actors and Training Diversity

Each training episode is populated with dynamic actors to prevent the policy from memorising a single static configuration. A single NPC patrol vehicle is driven along waypoint loops through the lot aisles by a custom proportional heading controller, and obstacle avoidance logic is incorporated: a forward-cone check brakes the vehicle when another actor is detected within a configurable stop distance.

Within each designated pedestrian zone, a single pedestrian may be spawned with a configurable probability per episode; spawned pedestrians execute random walks via `WalkerControl` with zone confinement and ego-avoidance logic. Bay occupancy is randomised per episode, with the target bay and its immediate neighbours always kept empty for manoeuvring clearance. The lot perimeter is delineated by traffic cones at regular spacing, and collision with a perimeter cone constitutes an episode termination with a collision penalty.

5.2 Known Limitations and Future Directions

Several scope boundaries have been identified. The GNSS noise model currently applies isotropic diagonal covariance to the `NavSatFix` message; in practice, satellite geometry and multipath effects produce anisotropic error ellipses that vary with the vehicle’s position relative to surrounding structures,

and extending the model to incorporate directional covariance profiles is being considered.

The Markov chain transition matrix governing GNSS fix-state changes is hand-tuned to approximate physically plausible dwell times; refining these probabilities based on empirical data could improve training fidelity. As noted in Section 4, the adoption of RTK-GNSS restricts the system to outdoor parking lots with sky visibility; indoor or covered environments would require a different localisation source, though the uncertainty-conditioned policy architecture itself would transfer without modification. These boundaries do not affect the validity of the proposed architecture but represent directions along which the work could be extended.

6. CONCLUSION

The localisation pipeline, simulation environment, GNSS Markov noise model, layout generation scripts, inspection tools, live visualiser and evidential PPO implementation have been completed. Environment integration testing is being finalised as of mid-April 2026, with hyperparameter optimisation and ablation training scheduled to commence thereafter. The Gantt chart on the final page presents the timeline for the remaining dissertation phases.

Acknowledgments

This work is supervised by Dr Josef Bajada at the University of Malta. The Anthropic Claude AI system was used for code review and document drafting assistance.

7. REFERENCES

- [1] E. Yurtsever, J. Lambert, A. Carballo, and K. Takeda. A survey of autonomous driving: Common practices and emerging technologies. *IEEE Access*, 8:58443–58469, 2020.
- [2] S. Thrun, W. Burgard, and D. Fox. *Probabilistic Robotics*. MIT Press, Cambridge, MA, 2005.
- [3] International Organization for Standardization. ISO 26262:2018 road vehicles – functional safety, 2018.
- [4] A. Kendall and Y. Gal. What uncertainties do we need in Bayesian deep learning for computer vision? *arXiv preprint arXiv:1703.04977*, 2017.
- [5] B. Charpentier, R. Senanayake, M. Kochenderfer, and S. Günnemann. Disentangling epistemic and aleatoric uncertainty in reinforcement learning. *arXiv preprint arXiv:2206.01558*, 2022.
- [6] A. Akgül, G. Baykal, M. Haußmann, and M. Kandemir. Overcoming non-stationary dynamics with evidential proximal policy optimization. *Trans. Mach. Learn. Res. (TMLR)*, 2025.
- [7] H. Ham and H. Ahn. DRO-EDL-MPC: Evidential deep learning-based distributionally robust model predictive control for safe autonomous driving. *arXiv preprint arXiv:2507.05710*, 2025.
- [8] A. C. Stutts, D. Erricolo, T. Tulabandhula, and A. R. Trivedi. Echoes of Socratic doubt: Embracing uncertainty in calibrated evidential reinforcement learning. *arXiv preprint arXiv:2402.07107*, 2024.
- [9] G. Puthumanai, P. Padrao, J. Fuentes, L. Bobadilla, and M. Ornik. GUIDED agents: Enhancing navigation policies through task-specific uncertainty abstraction in localization-limited environments. *IEEE Robot. Autom. Lett.*, 2025.
- [10] J. A. Placed and J. A. Castellanos. A deep reinforcement learning approach for active SLAM. *Appl. Sci.*, 10(23):8386, 2020.
- [11] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun. CARLA: An open urban driving simulator. In *Proc. 1st Annu. Conf. Robot Learn. (CoRL)*, volume 78 of *PMLR*, pages 1–16, 2017.
- [12] T. Moore and D. Stouch. A generalized extended Kalman filter implementation for the Robot Operating System. In *Proc. Intell. Auton. Syst. 13 (IAS-13)*, volume 302 of *Adv. Intell. Syst. Comput.*, pages 335–348. Springer, 2014.
- [13] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [14] A. Amini, W. Schwarting, A. Soleimany, and D. Rus. Deep evidential regression. In *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, volume 33, pages 14927–14937, 2020.
- [15] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 2nd edition, 2018.
- [16] J. Schulman, S. Levine, P. Moritz, M. I. Jordan, and P. Abbeel. Trust region policy optimization. In *Proc. 32nd Int. Conf. Mach. Learn. (ICML)*, volume 37 of *PMLR*, pages 1889–1897, 2015.
- [17] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann. Stable-Baselines3: Reliable reinforcement learning implementations. *J. Mach. Learn. Res.*, 22(268):1–8, 2021.
- [18] M. Sensoy, L. Kaplan, and M. Kandemir. Evidential deep learning to quantify classification uncertainty. In *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, volume 31, pages 3183–3193, 2018.
- [19] J. Gao, M. Chen, L. Xiang, and C. Xu. A comprehensive survey on evidential deep learning and its applications. *arXiv preprint arXiv:2409.04720*, 2024.
- [20] D. Oh and B. Shin. Improving evidential deep learning via multi-task learning. In *Proc. AAAI Conf. Artif. Intell.*, volume 36, pages 7895–7903, 2022.
- [21] B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, volume 30, 2017.
- [22] Z. Zhang, Q. Liu, Y. Li, K. Lin, and L. Li. Safe reinforcement learning in autonomous driving with epistemic uncertainty estimation. *IEEE Trans. Intell. Transp. Syst.*, 25(10):13653–13666, 2024.
- [23] Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proc. 33rd Int. Conf. Mach. Learn. (ICML)*, pages 1050–1059, 2016.
- [24] W. R. Clements, B. M. Robaglia, B. van Delft, R. B. Slaoui, and S. Toth. Estimating risk and uncertainty in deep reinforcement learning. *arXiv preprint arXiv:1905.09638*, 2019.
- [25] C. Kim, J.-K. Cho, H.-S. Yoon, S.-W. Seo, and S.-W. Kim. UNICON: Uncertainty-conditioned policy for robust behavior in unfamiliar scenarios. *IEEE Trans. Neural Netw. Learn. Syst.*, 2022.
- [26] C.-J. Hoel, K. Wolff, and L. Laine. Ensemble quantile networks: Uncertainty-aware reinforcement learning with applications in autonomous driving. In *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, pages 3440–3447, 2023.
- [27] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, volume 80 of *PMLR*, pages 1861–1870, 2018.
- [28] I. Kovacs, M. Stefanoni, R. Pesti, D. Csik, P. Sarcevic, and A. Odry. Evaluation of optimal covariance models for EKF-based wheeled mobile robot localization. *Acta Polytechnica Hungarica*, 22:143–158, 2025.
- [29] Z. Yang, Z. Peng, and J. Ma. SEG-Parking: Towards safe, efficient, and generalizable autonomous parking via end-to-end offline reinforcement learning. *arXiv preprint arXiv:2509.13956*, 2025.
- [30] F. Codevilla, A. M. López, V. Koltun, and A. Dosovitskiy. On offline evaluation of vision-based driving models. *arXiv preprint arXiv:1809.04843*, 2018.
- [31] W. Hess, D. Kohler, H. Rapp, and D. Andor. Real-time loop closure in 2D LIDAR SLAM. In *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, pages 1271–1278, 2016.
- [32] A. Y. Ng, D. Harada, and S. J. Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proc. 16th Int. Conf. Mach. Learn. (ICML)*, pages 278–287, 1999.

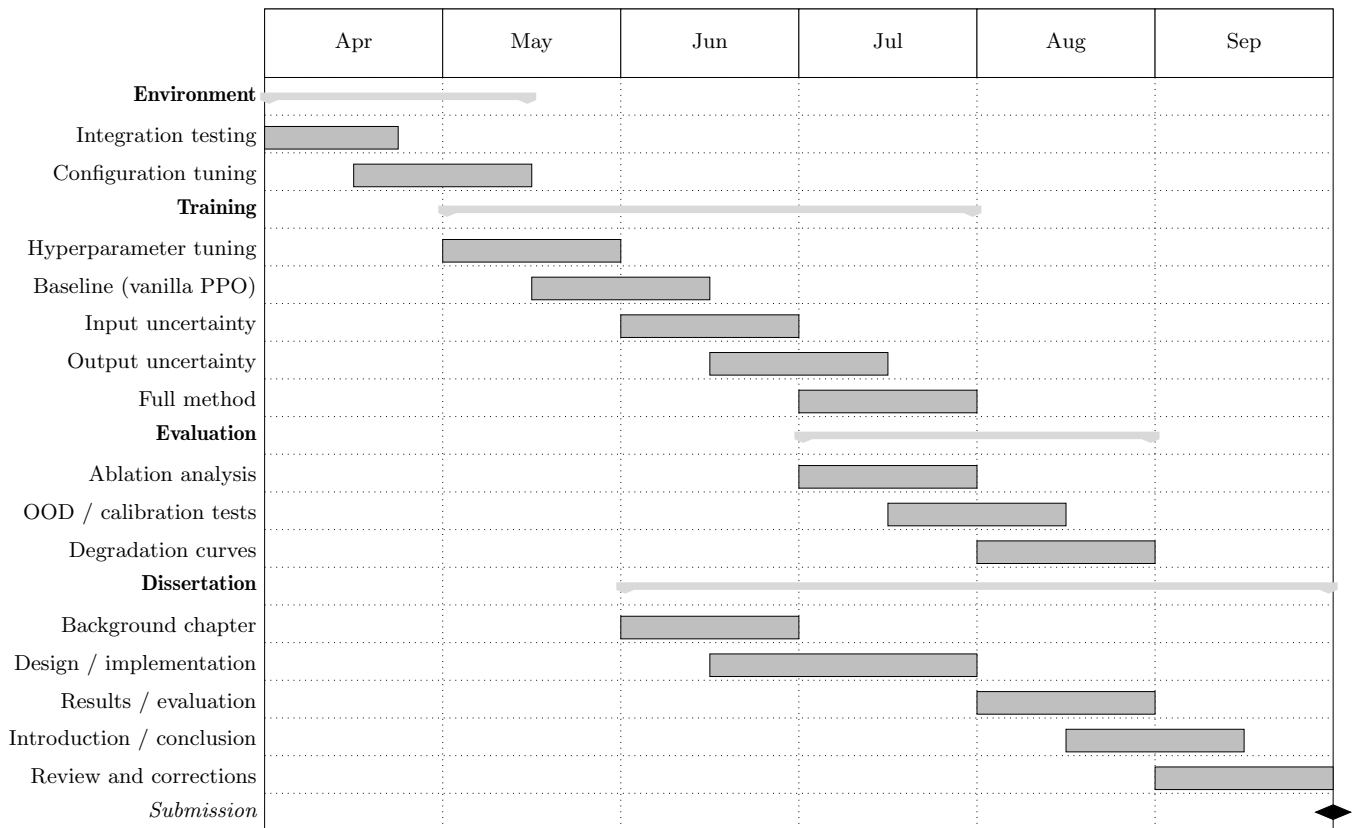


Figure 6: Dissertation work plan (April–September 2026).